

Deepak Mallampati

FORWARD DEPLOYED ENGINEER - APPLIED AI & AGENTIC SYSTEMS

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PROFESSIONAL SUMMARY

Applied AI / Forward Deployed Engineer with 4+ years shipping production Generative AI for regulated, security-sensitive enterprises. Build and deploy customer-facing RAG and agentic applications in Python end-to-end - dense retrieval with cross-encoder re-ranking (92% precision, -40% hallucination over millions of documents) and conductor-subagent agentic orchestration (-42% tokens). Deep experience deploying AI over sensitive data with privacy-preserving and compliance controls (HIPAA/SOC2/PCI, PII redaction, prompt-injection mitigation), partnering directly across 20+ teams to take solutions from concept to production.

CORE COMPETENCIES

Production Python • RAG Pipelines • Agentic Systems • Customer-Facing Delivery • Private-Cloud / On-Prem Deployment • Data Security & Privacy • Full-Stack Engineering • LLM Evaluation

Generative AI & LLMs: GPT-4o/GPT-5, Claude, Gemini, LLaMA | LangChain/LangGraph | RAG | Multi-Agent Systems | Prompt Engineering | Fine-tuning (LoRA/QLoRA/PEFT) | Hugging Face

Retrieval & Eval: FAISS, Pinecone, Weaviate | Cross-Encoder Re-ranking | Hybrid Search | RAGAS, BERTScore

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda), Azure OpenAI, GCP Vertex | Docker, Kubernetes, Terraform | CI/CD | MLflow, W&B

Security & Compliance: HIPAA, SOC 2, PCI DSS | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging | Zero-Trust

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira, Confluence

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Led end-to-end **deployment** of AI services with privacy-preserving data workflows (k-anonymity, l-diversity, differential privacy) enabling compliant analytics over sensitive data with zero PII exposure.
- Built fault-tolerant **automation infrastructure** (scheduled orchestration + exponential-backoff retry), reducing failed production runs by **60%**; established an LLM eval framework (RAGAS, BERTScore) with latency/accuracy dashboards.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ agents**, integrating with production data pipelines and observability tooling.
- Optimized prompts and evaluated GPT-4o, Claude, and Gemini, improving task success rates by **27%**; engineered safeguards reducing unsafe outputs by **42%**.
- Built **12 APIs and microservices**, cutting p95 latency 1.5s → **800ms**; partnered across **20+ internal teams** serving **100,000+ requests/day**.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, Laplace differential privacy), cutting re-identification risk from **3.38%** → **0.32%** while retaining **99%** of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an H100 for controllable long-form generation; authored an IEEE-format conference paper.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Trained regression/classification models for equipment-failure prediction and anomaly detection; fine-tuned BERT/RoBERTa for NER achieving **89% F1**.

PROJECTS

career-ops

An autonomous, AI-driven pipeline that scans, scores, and generates tailored applications overnight without supervision - end-to-end ownership of a real deployed system.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - solo concept-to-production delivery.

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive data without exposing identities - quantifying the privacy/utility trade-off.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0